

Erdős Problem #1002

EP-1002

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

For any $0 < \alpha < 1$, let $f(\alpha, n) = \frac{1}{\log n} \sum_{1 \leq k \leq n} (\frac{1}{2} - \{\alpha k\})$. Does $f(\alpha, n)$ have an asymptotic distribution function?

In other words, is there a non-decreasing function g such that $g(-\infty) = 0$, $g(\infty) = 1$, and $\lim_{n \rightarrow \infty} |\{\alpha \in (0, 1) : f(\alpha, n) \leq c\}| = g(c)$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>